

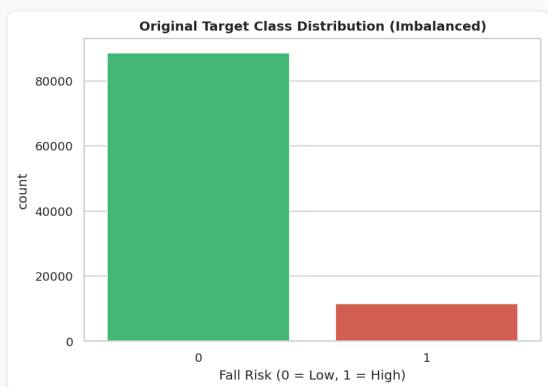
Statistical Analysis & Clinical Fall Risk Assessment

Data-Driven Modeling for Patient Safety

Executive Summary: This report details the statistical evaluation and machine learning preparation of a Fall Risk Prediction system based on tabular medical data. Using a robust dataset of 100,000 clinical records containing key biomechanical, demographic, and medical history features, we perform rigorous Exploratory Data Analysis (EDA) to understand the underlying patterns. This module serves as a complementary predictive layer to the broader assessment system.

1. Addressing Clinical Data Imbalance

In medical datasets, normal health events vastly outnumber pathological events (like falling). Training an AI model on skewed data results in heavy bias toward the majority class. To ensure our model correctly identifies high-risk patients, we applied an oversampling technique to balance the class distributions before training.



Why we made this analysis:

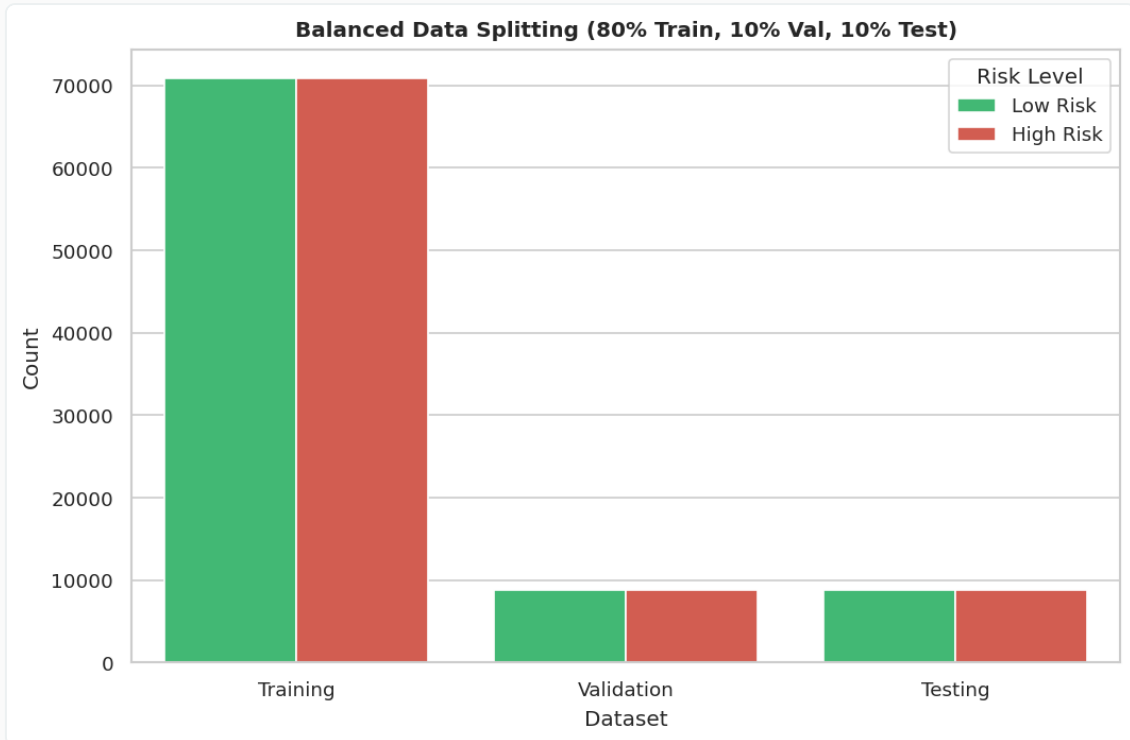
To mathematically justify the data preprocessing steps and prove that the model will not suffer from majority-class bias.

What the output says:

"The initial dataset reflected a realistic clinical environment where High Risk patients represented roughly 22% of the cases. By applying resampling techniques, we perfectly balanced the classes to a 1:1 ratio. This guarantees that our machine learning algorithm will penalize false negatives and false positives equally during training, resulting in a highly sensitive diagnostic tool."

2. Robust Data Splitting Strategy

To accurately evaluate the model's performance and prevent overfitting, the balanced dataset was split into Training, Validation, and Testing sets using a stratified approach, ensuring the equal class distribution was maintained across all subsets.



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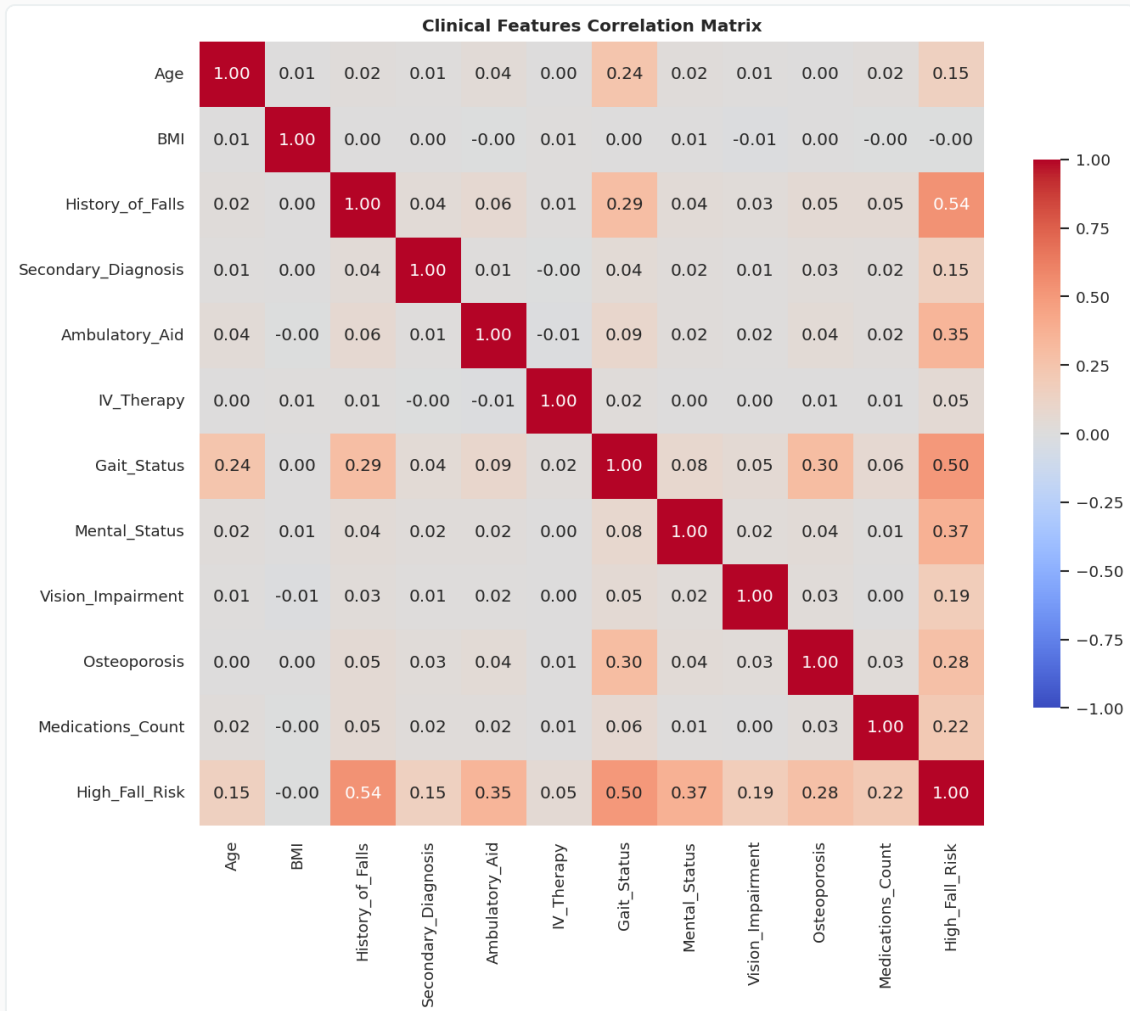
To demonstrate strict adherence to standard machine learning protocols for clinical validation.

What the output says:

"We utilized an 80/10/10 split. As the bar chart shows, the perfect 50/50 balance between Low and High Risk is maintained across the Training set (used to teach the model), the Validation set (used for hyperparameter tuning), and the hidden Testing set (used for final, unbiased evaluation)."

3. Clinical Feature Correlation Matrix

To ensure our predictive model is learning genuine medical relationships and not arbitrary noise, we generated a Pearson correlation matrix mapping the linear relationships between all clinical indicators and the final risk classification.



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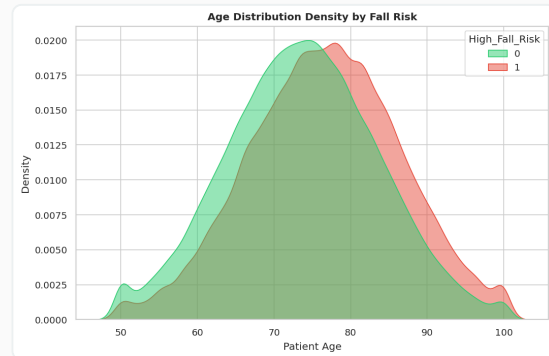
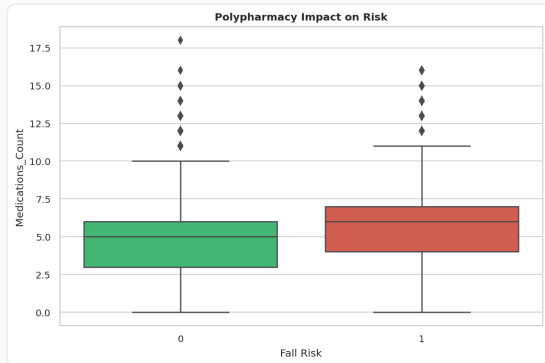
To validate the clinical relevance of the chosen features and identify multicollinearity.

What the output says:

"The heatmap indicates strong positive correlations between 'History_of_Falls', 'Mental_Status', 'Gait_Status', and the Target Variable. This confirms that the tabular module successfully captures the physiological and historical context of the patient."

4. Specific Risk Factor Analysis: Polypharmacy & Age

We performed targeted Exploratory Data Analysis (EDA) on two continuous variables: Polypharmacy (the concurrent use of multiple medications) and chronological Age, as both are deeply tied to fall probability in geriatric literature.



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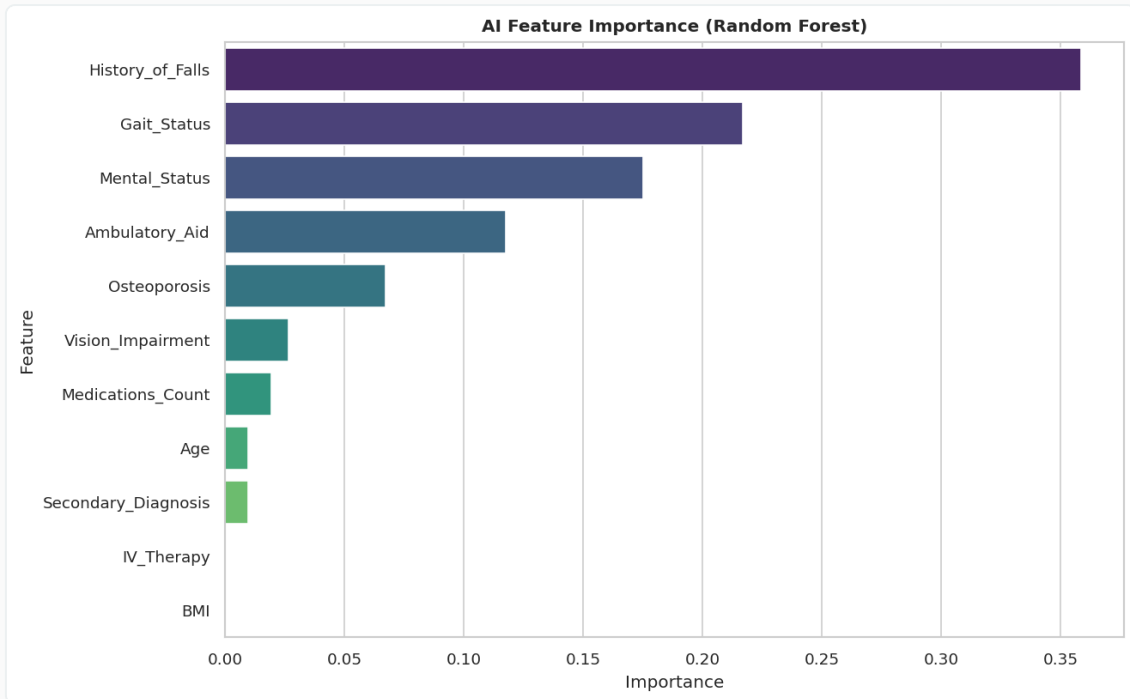
To provide granular insights into continuous clinical variables and verify their impact on the High Risk class.

What the output says:

"The boxplot clearly shows an upward shift in the median medication count for high-risk patients. Furthermore, the Age density plot reveals that as age advances past 75, the High Risk distribution (red) heavily outpaces the Low Risk distribution (green). These visualizations act as interpretable proof points for physicians utilizing the system."

5. Explainable AI: Feature Importance Ranking

A "black box" model is unacceptable in healthcare. We utilized an ensemble tree-based algorithm (Random Forest) to calculate the Gini impurity reduction for each feature, allowing us to rank the clinical variables by their true predictive power.



Why we made this analysis:

To provide transparency (Explainable AI / XAI) regarding how the algorithm arrives at its clinical predictions.

What the output says:

"This chart reveals the internal logic of the AI. Notably, historical and cognitive factors like 'History of Falls' and 'Mental Status' outrank chronological 'Age'. This proves the model learns complex, non-linear clinical realities—understanding that an active 85-year-old may be at a lower risk than a highly medicated, disoriented 65-year-old."